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Comparative analysis of hydrogen/air combustion CFD-modeling for 3D and 2D computational domain of micro-cylindrical combustor

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ABSTRACT

A computational study of a micro-cylindrical premixed hydrogen-air combustor using RANS is presented. The influence of the geometric dimensionality of the computational domain is studied by performing numerical simulations with 2D planar, 2D axisymmetric and 3D meshes. A commercial software package with a particular choice of several physical models (regarding turbulence, combustion, radiation) is used for this analysis. A mathematical model describing the hydrogen/air combustion in a cylindrical combustor is developed. The reaction scheme of combustion process with 9 species and 19 steps was simulated by usage Eddy Dissipation Concept (EDC) model. The obtained results from presented CFD model were compared with experimental and numerical data of other publications. The combustion characteristics of developed numerical model such as wall and fluid temperature, hydrogen mass fraction, velocity and pressure were investigated for different geometry types. It is established that the 2D approach to solving combustion problems leads to significant deviations of the obtained results from real ones and can be used only for a preliminary evaluation of the characteristics of the combustion process. Based on the developed model, it is established that the difference between the combustion products temperature at the combustor outlet for the three- and two-dimensional (2D-planar and 2D-axis) geometries is more than 25%, and for the range of the flame formation the temperatures differ by several times. The contours of pressure have minimal visual differences for the all hydrogen burning characteristics that are investigated in this work. The axial pressure profiles for all investigated geometries (2D-axis, 2D-planar and 3D) are almost similar in terms of trend and value.

Introduction

Numerical studies of various thermal, physical, chemical and other processes are increasingly used in the work of engineers and scientists in different countries. According to the NASA (National Aeronautics and Space Administration) report [1–3], approximately 85% of the total computing power of giant-powered computer in all over the world is used to solve problems of computational fluid dynamics (CFD). In addition, the same organization presented two reports of Revolutionary Computational Aerosciences [4] and Vision 2030 [5], which show the prospects for using numerical modeling of various physical and chemical processes. Thus, the Vision 2030 report

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indicates that the estimated computing capacities for solving the problems of computational fluid dynamics will increase by approximately 10–12% per year in the next two decades. The main advantage of numerical modeling (CFD modeling) is the possibility of obtaining visual information about the nature of the investigated processes without the use of expensive experimental equipment.

Combustion of different hydrocarbon fuels in cylindrical combustors, internal combustion engines, gas turbines is one of long-term tasks of CFD [6–9]. The modern tools of computational fluid dynamics allow to predict the performance of different combustion devices mentioned above and combustion parameter of various fuels. The number of scientific publications devoted to the numerical study of combustion processes has been steadily increasing in recent years. There is a large number of publications devoted to the numerical study of combustion processes of methane [10–13], hydrogen [14–17], synthesis gas [18,19], etc.

Numerical investigations of hydrogen/air flames have a high demand for CFD application. For CFD modeling of hydrocarbon fuels combustion, specialized software such as ANSYS (CFX and Fluent) [20–22], Comsol Multiphysics [23], OpenFOAM [24,25], etc. are used as well as specially written codes in various programming languages [26–29]. Hua and co-workers [30] presented modeling investigation of premixed hydrogen combustion for various dimensions of the combustion chamber at constant excess air factor. They numerically analyzed the influence of different initial conditions on heat transfer at combustion chamber wall. Also they concluded that these conditions have a profound effect on hydrogen/air flame. The 2D geometry of computational domain was used for that CFD investigation. Li and co-authors [31] carried out a numerical investigation of hydrogen/air premixed combustion in the microcylindrical combustor. For these purposes the two-dimension governing equations (mass conservation, energy and species transport, momentum, energy conservation for steel wall) are solved by using ANSYS Fluent. They also used 2D geometry of combustion chamber in which hydrogen/air combustion is obtained. Yang et al. [32] built a mathematical model to determine the effect of pressure decreasing on hydrogen/air combustion efficiency in the micro-combustion chamber. They analyzed a 2D geometry of the micro-combustor with dual cavities. The software Fluent 6.3 was used for numerical study.

Numerical study of hydrogen and other hydrocarbon fuels combustion requires considerable computer processing power. Therefore, the 2D computational domain is often simplified to an axisymmetric one. Roy et al. [33] presented an simulation study of methanol and synthesis gas bluff-body flames by using Reynolds-averaged Navier-Stokes based on different turbulence models. For this aim, they developed two-dimension axisymmetric geometry. Axis boundary condition was provided along the x-axis. The mathematical model was solved via ANSYS Fluent CFD code.

The use of three-dimensional computational domain has recently become more widely used, due to the increase of computer power. Yilmaz and co-workers [34] presented a simulation investigations of hydrogen-air combustion process to understand the effect of different turbulence models (k-ε (standard, RNG and realizable) and Reynolds Stress Model – RSM) on combustion and emission characteristics. For these investigations 3D model of combustion chamber was developed and numerically solved via ANSYS Fluent software. The mesh structure was contained about 2.5 million elements. They reported that the choice of the turbulence model has a significant effect on the result of the study. The obtained results by RNG k-ε turbulence model have well agreement with experimental results, when RNS turbulence model was not able to predict detailed characteristics of hydrogen/air flame.

In addition, Jiaqiang et al. [35] developed a tree-dimension numerical model to understand effects of inlet pressure on premixed hydrogen-air combustion in a micro-cylindrical combustor with a step. They determined the effect of mesh structure on numerical results of combustion characteristics. A comparison of wall temperature that is calculated for 1,669,704, 557,574 and 2,230,274 was presented. To verify the developed model, the wall temperature of the combustion chamber was compared with the experimental data. Agreement between the numerical results and experimental data was obvious.

Research studies on numerical investigation of premixed hydrogen/air combustion are mainly focused on study of combustion characteristics only for one computational domain, for example, only for 2D or only for 3D geometry. However, the results of the numerical study are influenced by the nature of the calculated computational geometry [36]. The discrepancy between the calculation results for 3D, 2D and 2D-axisymmetric geometry can be significant [37,38]. However; there is no literature (according author) on comparison study of 2D planar, 2D axis and 3D numerical investigations on hydrogen/air combustion characteristics. Whether numerical study will partly or fully replace experimental investigations will significant depend on the reliability of the combustion and turbulence models used and a comprehensive analysis of the obtained results. Therefore, the study of the influence of the computational domain structure on combustion characteristics is a crucial task.

In this paper, a comparison study of three-dimensional (3D), two-dimensional (2D) and two-dimensional axisymmetric (2D-axis) numerical investigation of hydrogen-air premixed combustion is performed. The research was carried out in the ANSYS Fluent software. Based on the obtained results, the influence of the chosen computational domain on the distribution of temperature, hydrogen mass fraction, velocity and pressure of the combustion products was determined.

Numerical setup

Computational domain

To understand the effect of geometry on the hydrogen combustion characteristics, three computational domain of the combustor are considered: two-dimensional axisymmetric (a), two-dimensional (b), three-dimensional (c) as shown in Fig. 1.

The schematic diagram of combustion chamber and all dimensions are shown in Fig. 2. The total length of combustor is 18.0 mm. The inlet and outlet diameters are 2.0 and 3.0 mm, respectively. The outer diameter of combustion chamber is 4.0 mm. The material of wall is steel with the corresponding properties [35].

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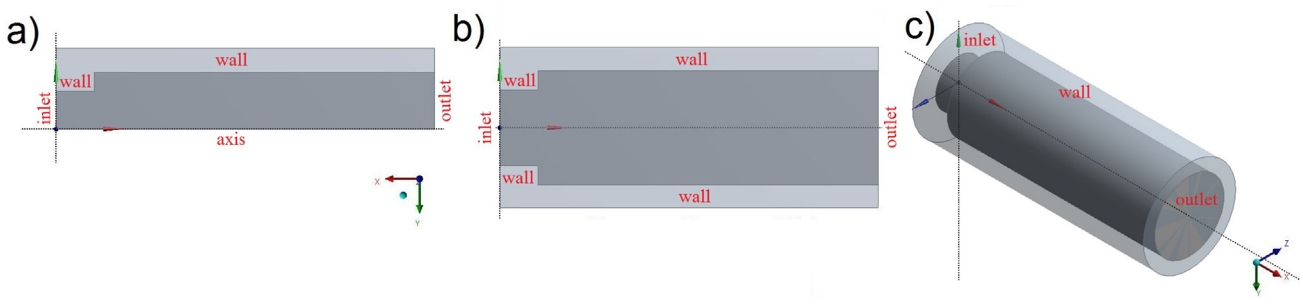

Fig. 1 – The computational domain of combustion chamber: a – 2D-axis; b – 2D; c – 3D.

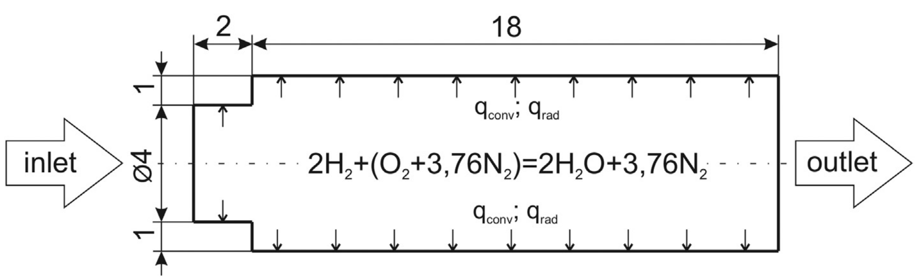

Fig. 2 – The schematic diagram of hydrogen/air 2D combustor and dimensions of it.

Mathematical model

In this paper, H_2 burned as gas fuel at stoichiometric conditions, i.e. excess air ratio is equal to unit. In developed hydrogen combustion model the conservation equation, momentum and energy equation in combination with species equations are solved in ANSYS Fluent. The heat transport and body force caused by concentration gradients can be neglected due to its low value. Moreover, other assumptions are taken for CFD-modeling: steady-state conditions of combustion; no the flux of energy due to a mass concentration gradient (no Duflor effects) [39]; work by viscous forces and by pressure is not done; surface oxidation reactions of the metal wall are absent. Combustion reaction scheme with 9 species and 19 steps was simulated by usage EDC model (Eddy Dissipation Concept) [40].

Continuity equation for steady-state conditions:

$$\nabla\cdot(\rho\vec{v})=0 \tag{1}$$

For 2D-axis geometry, the equation for conservation of mass can be written as follows:

$$\frac{\partial}{\partial x}(\rho v_x)+\frac{\partial}{\partial r}(\rho v_r)+\frac{\rho v_r}{r}=0 \tag{2}$$

In these equations v_x and v_r are the axial and radial velocity, respectively; x and r are the axial and radial coordinate, respectively.

Conservation of momentum in an inertial (non-accelerating) reference frame is described by follow equations [41,42]:

$$\nabla\cdot(\rho\vec{v}\vec{v})=-\nabla\cdot p+\nabla\cdot(\overline{\overline{\tau}})+\rho\vec{g}+\vec{F} \tag{3}$$

where p is the static pressure; $\rho\vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively. The stress tensor $\overline{\overline{\tau}}$ is given by follows equation:

$$\overline{\overline{\tau}}=\mu\left[\left(\nabla\vec{v}+\vec{v}^{T}\right)-\frac{2}{3}\nabla\cdot\vec{v}I\right] \tag{4}$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The radial and axial momentum conservation equations for 2D-axis geometry can be found in Ref. [41].

The conservation equation of energy for combustion products can be written as follows:

$$\nabla\cdot\vec{u}(p+\rho\cdot E_f)=\nabla\cdot\left[k_{eff}\cdot\nabla T-\left(\sum h_j\cdot\vec{J}_j\right)+\overline{\overline{\tau}}\cdot\vec{v}\right]+S_f^h \tag{5}$$

In this equation T is the combustion product temperature; ρ is the density; E_f is total fluid energy; $\vec{J}_j$ is the diffusion flux of j element; k_{eff} is the conductivity; h_j is the enthalpy of j element; S_f^h is the fluid enthalpy source term (include the source of energy due to chemical reaction).

Conservation equation of energy for steel wall:

$$\nabla\cdot(k_w\cdot\nabla T)=0 \tag{6}$$

where k_w is the thermal conductivity of the steel wall.

Transport of combustion products is described by follow equations:

$$\nabla\cdot(\rho\cdot\vec{v}\cdot Y_j)=-\nabla\vec{J}_j+R_j \tag{7}$$

where R_j is chemical reaction rate of species j formation or decomposition; Y_j is the mass fraction of j element of combustion products; $\vec{J}_j$ is diffusion flux of j species.

Diffusion flux of j species can be written as:

$$\vec{J}_j = -\left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t}\nabla Y_j - D_{T,j}\frac{\nabla T}{T}\right) \tag{8}$$

where $D_{j,m}$ and $D_{T,j}$ are the mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and μ_t are the turbulent Schmidt number and turbulent viscosity, respectively.

Chemical reaction rate R_i can be calculated by Eddy-Dissipation Concept (EDC) chemical reaction model. The hydrogen-air flame is simulated by a detailed chemical mechanism of 9 elements and 19 chemical reversible reactions [43].

According hydrogen-air combustion study [34] RNG k-ε model has the best agreement with experimental data. In present investigation Re > 1000, therefore the k-ε turbulence model can be properly chosen [44].

$$\frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j}\left(\alpha_k \mu_{eff}\frac{\partial k}{\partial x_j}\right) + G_k + G_b - \rho\varepsilon - Y_M + S_k \tag{9}$$

and

$$\frac{\partial}{\partial x_i}(\rho\varepsilon v_i) = \frac{\partial}{\partial x_j}\left(\alpha_\varepsilon \mu_{eff}\frac{\partial \varepsilon}{\partial x_j}\right) + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon}G_b) - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} - R_\varepsilon + S_\varepsilon \tag{10}$$

In equations for k-ε turbulence model, G_k shows the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. The quantities α_k and α_ε are the inverse effective Prandtl numbers for k and ε, respectively. S_k and S_ε are user-defined source terms.

The additional term R_ε in the ε equation can be determined as follows:

$$R_\varepsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0)}{1 + \beta\eta^3}\frac{\varepsilon^2}{k} \tag{11}$$

where $\eta \equiv Sk/\varepsilon$, $\eta_0 = 4.38$, $\beta = 0.012$.

The model constants $C_{1\varepsilon}$ and $C_{2\varepsilon}$ in k-ε turbulence model used by default setup for Fluent solver of ANSYS:

$$C_{1\varepsilon} = 1.42, \quad C_{2\varepsilon} = 1.68 \tag{12}$$

Eddy Dissipation Concept model

Turbulence-chemistry interactions have been simulated with the Eddy Dissipation Concept (EDC) model. Fast chemistry approaches (such as Flamelet) were not used as it is recognized that moderate or intense low-oxygen dilution (MILD) combustion requires finite rate chemistry models [45,46]. The Eddy Dissipation Concept (EDC) developed by Magnussen and Hjertager [47] was used to model detailed chemistry. The EDC model has the advantage of incorporating the effect of finite rate kinetics at computing costs which are quite moderate compared to more advanced models as the transported PDF method. According to Eddy Dissipation Concept, hydrogen combustion performs in the areas of the flow where the dissipation of turbulent kinetic energy occurs. Such areas are defined as fine structures and they can be characterize as perfectly stirred chemical reactors. The mean residence time of the fluid within the fine structures (τ^*) and the mass fraction of the fine structures (γ_λ) are provided by an energy cascade model, which describes the energy dissipation process as a function of the characteristic scales:

$$\tau^* = 0.41\left(\frac{\nu}{\varepsilon}\right)^{0.5} \tag{13}$$

and

$$\gamma_\lambda = 2.13\left(\frac{\nu \cdot \varepsilon}{k^2}\right)^{0.25} \tag{14}$$

where ε is the dissipation of turbulent kinetic energy, k; and ν is the kinematic viscosity.

Assuming that combustion obtain only in the fine structure, the reaction rates for formation and decomposition of all elements are determined from a mass balance of the fine structure of combustor and the net mean element reaction rate is modeled as:

$$\overline{\omega}_i = \frac{\overline{\rho}\cdot(\gamma_\lambda)^2}{\tau^*}\cdot\frac{Y_i^* - \tilde{Y}_i}{1 - \gamma_\lambda^3} \tag{15}$$

where Y_i^* is mass fraction of ith element in the fine structures; $\tilde{Y}_i$ is the mean mass fraction of the ith element between the fine structures and the surrounding area; $\overline{\rho}$ is the average density of the reaction flow. The mean mass fraction $\tilde{Y}_i$ can be determined by following expression:

$$\tilde{Y}_i = \gamma_\lambda^3 \cdot Y_i^* + (1 - \gamma_\lambda^3)\cdot Y_i^0 \tag{16}$$

The evolution of Y_i^* after reacting over time, depends on the chemical mechanism, which is chosen for computation. To reduce the computational time for integrating the Arrhenius reaction rates in these fine scales, the reaction scheme of combustion process with 9 species and 19 steps is used [48,49]. The reaction scheme of hydrogen/air combustion presented in Table 1.

Table 1 – Reaction Mechanism of hydrogen/air combustion.

No.	Reaction
1.	$H_2 + O_2 \Leftrightarrow 2OH$
2.	$OH + H_2 \Leftrightarrow H_2O + H$
3.	$H + O_2 \Leftrightarrow OH + O$
4.	$O + H_2 \Leftrightarrow OH + O$
5.	$H + O_2 \Leftrightarrow HO_2$
6.	$H + O_2 + O_2 \Leftrightarrow HO_2 + O_2$
7.	$H + O_2 + N_2 \Leftrightarrow H_2O + O_2$
8.	$OH + HO_2 \Leftrightarrow H_2O + O_2$
9.	$H + HO_2 \Leftrightarrow 2OH$
10.	$O + HO_2 \Leftrightarrow O_2 + OH$
11.	$2OH \Leftrightarrow O + H_2O$
12.	$H_2 \Leftrightarrow H + H$
13.	$O_2 \Leftrightarrow O + O$
14.	$H + OH \Leftrightarrow H_2O$
15.	$H + HO_2 \Leftrightarrow H_2 + O_2$
16.	$HO_2 + HO_2 \Leftrightarrow H_2O_2 + O_2$
17.	$H_2O_2 \Leftrightarrow OH + OH$
18.	$H_2O_2 + H \Leftrightarrow HO_2 + H_2$
19.	$H_2O_2 + OH \Leftrightarrow H_2O + HO_2$

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Combustion reaction occurs within a thin confined reaction area which is typically smaller than the size of the mesh element. Therefore, in EDC model the computational cell is divided into two sub-zones: the reacting zone fine structure (I zone) and the surrounding fluid (II zone) as shown in Fig. 3 All the homogeneous chemical reactions are assumed to occur in the fine structures that are locally treated as adiabatic, isobaric, Perfectly Stirred Reactors (PSR) transferring mass and energy only to the surrounding fluid.

Radiation model

Some authors suppose that for the modeling of hydrogen/air combustion it can be assumed that radiant heat exchange is not observed [50,51]. However, the expected combustion temperature can reach a level above 2000 K, therefore it is necessary to use the radiation model [52]. In tis study, P-1 radiation model is used for developed numerical model. The chosen P-1 radiation model is the simplest case of the more general P-N model, which takes into account the influence of geometry on the radiative heat transfer. For the absorbing, gray, scattering and also emitting medium containing emitting, absorbing, scattering particles, the transport equation for the incident radiation are defined as follows [53]:

$$\nabla\cdot(\Gamma\nabla G)+4\pi\left(a\frac{\sigma T^4}{\pi}+E_p\right)-(a+a_p)G=0 \tag{17}$$

In this equation Γ is the parameter of P-1 radiation model:

$$\Gamma=\frac{1}{(3(a+\sigma_s)-C\sigma_s)} \tag{18}$$

where σ_s is the scattering coefficient, a is the absorption coefficient of combustion products and wall, G is the incident radiation, C is the linear-anisotropic phase function coefficient.

In addition, in Eq. (17) the parameter E_p is the equivalent emission of the particles and a_p is the equivalent absorption coefficient depends on geometry characteristics. The mentioned values can be written as:

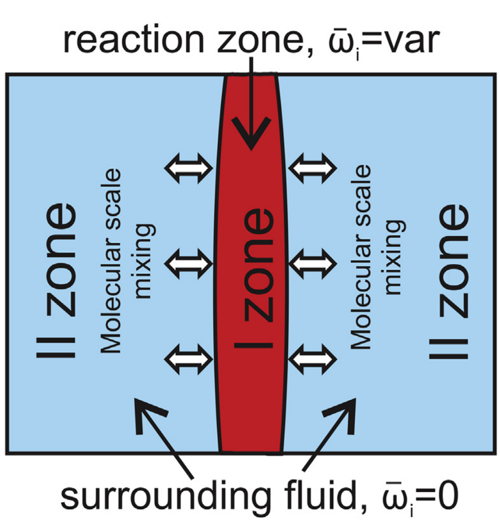

Fig. 3 – Schematic of computation cell based on EDC model.

$$E_p=\lim_{V\to 0}\sum_{n=1}^{N}\varepsilon_{pn}A_{pn}\frac{\sigma T_{pn}^4}{\pi V} \tag{19}$$

and

$$a_p=\lim_{V\to 0}\sum_{n=1}^{N}\varepsilon_{pn}\frac{A_{pn}}{V} \tag{20}$$

In these equations the summation is over N particles in volume V.

The main advantages of P-1 model:

- radiative transfer equation easy to solve with little CPU demand
- include effect of scattering
- work reasonably well for combustion application
- easy applied to complicated geometries with curvilinear coordinates.

Boundary conditions

The specific heat and density of the H_2-air inlet mixture are determined by the equations of mixing-law and incompressible-ideal-gas law [54], respectively. The viscosity of the gas flow is determined by Sutherland viscosity law. The thermal conductivity of the gas flow is computed as average mass fraction of each element. The pressure-based solver is used. The convergence criterion for momentum, continuity and species are defined as 10^{-3}, for energy less than 10^{-6}. For described equations the second-order upwind interpolation is used. The boundary conditions for the model are presented in Table 2.

Mesh

In solving problems of computational fluid dynamics, the accuracy of the results is determined not only by the mathematical description of physical and chemical processes and the convergence criteria, but also by the quantity and quality of the elementary cells of the computation mesh. Therefore, various meshes are used to determine effect of mesh element number on the obtained results. In accordance with the purposes of this study, the number of cells for two-dimensional and three-dimensional geometry was determined, in which a further increase in their number did not have a significant effect on the final results. Adaptation of the computational grid is not carried out because it is necessary to obtain an exact dependence of the influence of the cells number on the

Table 2 – Boundary conditions.

No.	Parameter	Inlet	Outlet
1	Excess air ratio	1.0	–
2	Mass flow rate H_2+air	1.8551×10^{-5} kg/s	–
3	H_2 mass flow rate	5.25×10^{-7} kg/s	–
4	Mass fraction: $H_2/O_2/N_2$	0.028301/0.22641/0.745289	–
5	Turbulent intensity	5%	5%
6	Hydraulic diameter	2×10^{-3} m	3×10^{-3} m
7	Gauge pressure	0.0 Pa	0.0 Pa

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final results. The computation mesh for 2D and 2D geometries is generated in the ANSYS Meshing. Six various meshes are used to determine the axial temperature in combustion chamber. The models that included these various grid structures are modeled under equal boundary conditions, which presented in Table 2. The profiles of axial temperature of combustion products are compared for 2D (Fig. 4) and 3D (Fig. 5) geometries.

For the results shown in Figs. 4 and 5, convergence of solutions in 350–510 iterations was observed. Based on the obtained temperature profiles, it can conclude the following: for 2D geometry, increasing the number of grid elements above 50,000 does not significantly effect on the final results; for 3D geometry, increasing the number of grid elements above 400,000 does not have a significant effect on the final results.

Therefore, the mesh structure with 46,400 and 373,390 cells is chosen for 2D and 3D geometries, respectively, to save computational time. For 2D-axis geometry the mesh structure with 36,244 cells is chosen.

Results and discussion

Model verification

To verify developed model, the calculated profiles of wall temperature for tree types of geometries are compared with experimental investigation of Wenming and co-workers [55] and modeling investigation of Jiaqiang and co-authors [35].

As shown in Fig. 6, discrepancies between numerical and experimental results are observed at area close to inlet of the combustion chamber. These discrepancies decreases with the increase of wall distance to outlet. The difference can be due to errors of metering equipment of experimental set-up and

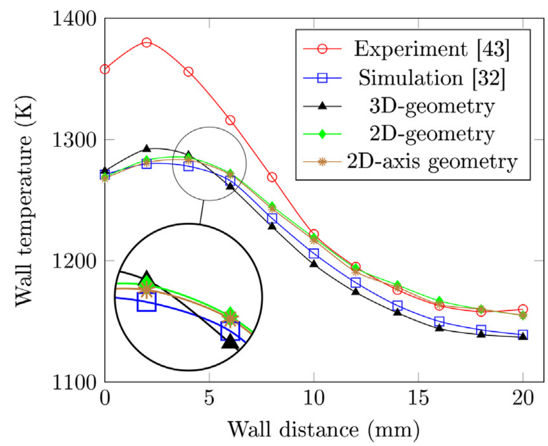

Fig. 6 – Comparison between numerical results of presented model with experimental [55] and simulation [35] results.

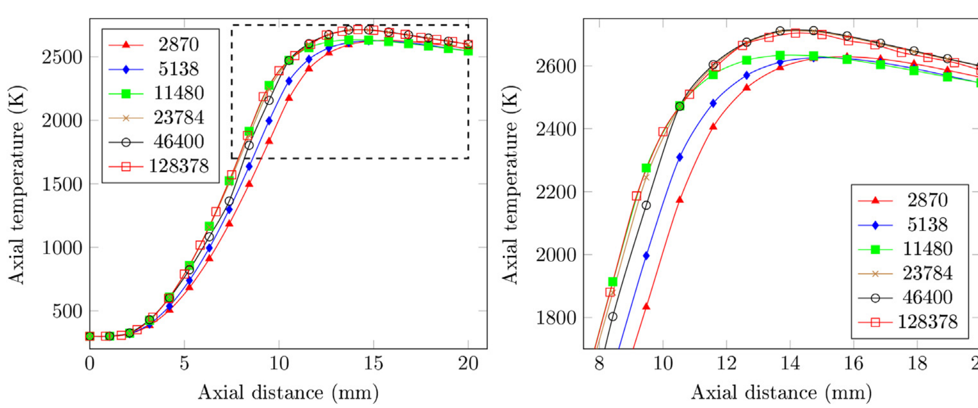

Fig. 4 – The combustion products temperature profile at different number of mesh elements for 2D geometry.

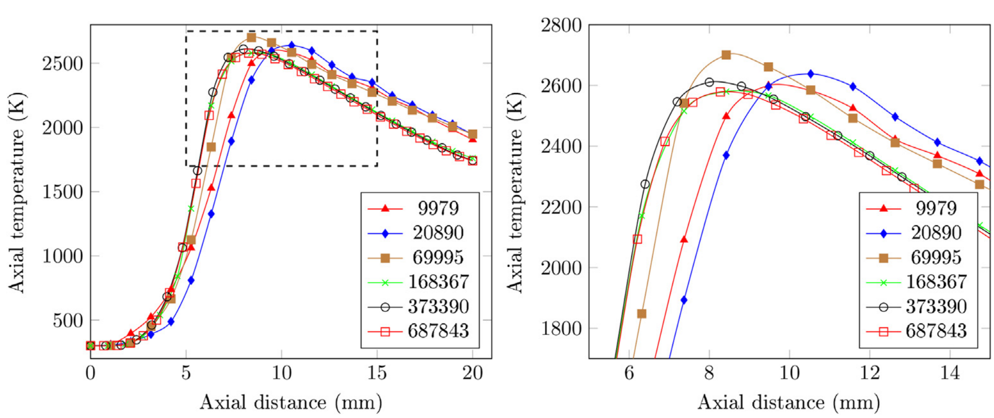

Fig. 5 – The combustion products temperature profile at different number of mesh elements for 3D geometry.

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assumptions for developed mathematical model. However, observed discrepancies reduce at the middle area of the combustion chamber and calculated profiles of wall temperature have a well agreement with experimental data in terms of trend and value. The maximum relative errors in area near inlet are 6.12% for 3D geometry, 7.11% for 2D geometry and 7.34% for 2D-axis geometry. Based on this, it can be concluded that developed mathematical model for three types of geometries is reliable to study the effects of different geometries on characteristics of hydrogen/air premixed combustion.

Effect of geometry on temperature contours

To evaluate the effect of the computational geometry on the temperature in the combustion chamber, the temperature contours were compared. Fig. 7 shows the temperature contours of the combustion products in the combustion chamber for 2D-axis (a), 2D (b) and 3D (c) geometries. Here, the contour for 2D-axis geometry is represented as a mirror image along the *x*-axis according Fig. 1. For three-dimensional geometry, the temperature contour for the XY-plane is also presented to better understand the effect of geometry. Numerical studies were performed for the initial conditions and convergence criteria presented in the Table 2. The settings for the Fluent solver were identical for all cases.

As shown in Fig. 7, the type of geometry has a significant effect on the temperature profile. Near the combustor inlet, the temperature contours for the investigated geometries are practically coincided. However, towards the outlet of the combustor, discrepancies become significant.

Such discrepancies can be explained by the following facts. Mixing and more generally diffusion phenomena, that takes place at hydrogen/air combustion, are never purely 2D. 3D diffusion is accounted for by the transport equation of turbulence quantities for the used RANS model and for laminar conditions every parameters depend on geometry of computational domain. For analyzed geometry the aspect ratio (Fig. 2; $x/y \approx 7$) is sufficiently small. (Here, *x* is combustor length; *y* is combustor diameter). For such aspect ratio, the 2D model cannot be considered as three-dimensional, because third dimension is not too much larger than any other dimensions. Only for large aspect ratios (above 50, [37]) there should not be much difference. And 2D modeling can be considered as 3D, in which the third dimension is much large than any other dimension.

Fig. 8 presents axial temperature profile for different computational geometries. It also can be seen that the near combustor inlet the temperature profiles have a well agreement between results for all investigated geometries, because the temperature in that area is lower. However, towards the

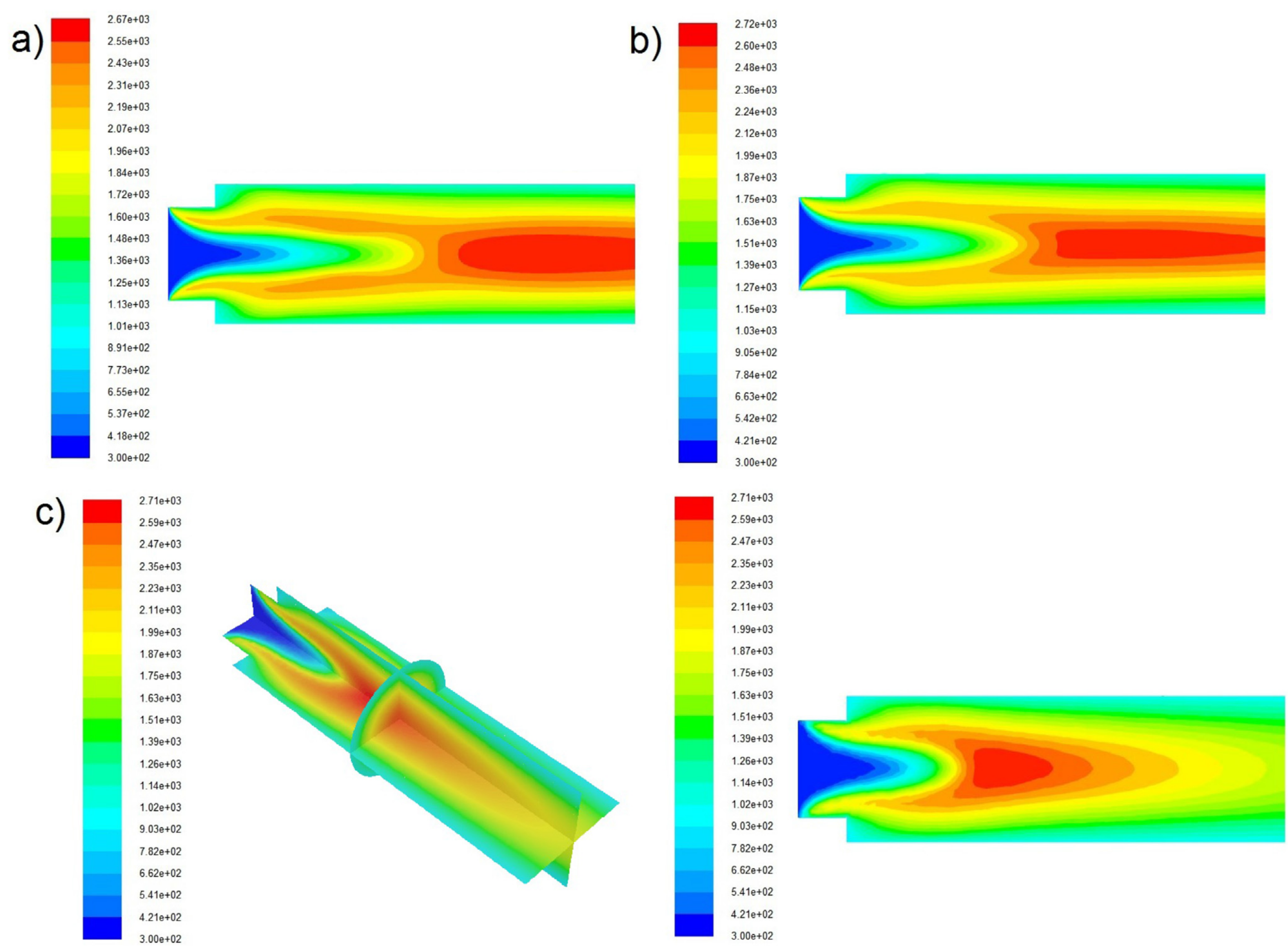

Fig. 7 – Contours of temperature in the combustor for different computational geometries: a) 2D-axis; b) 2D; c) 3D.

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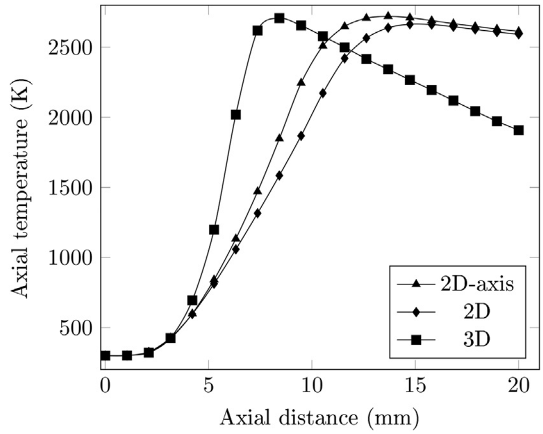

Fig. 8 – Axial temperature profiles in the combustor for different computational geometries.

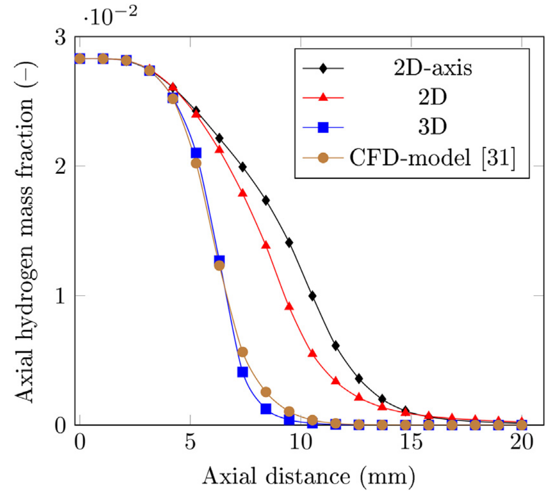

Fig. 10 – Comparison of axial H_2 mass fraction profiles in the combustor for different computational geometries with simulation results [34].

outlet of the combustor, discrepancies become significant. From Fig. 8, it can be seen that the difference between the temperature of the combustion products at the outlet for three-dimensional and 2D (2D and 2D-axis) geometries is more than 25%, and for the area of the flame formation, the temperatures differ by several times. Also for the area of the formation of the flame there are discrepancies between the results for 2D and 2D-axis geometries, provided the same dimension and structure of the computational grid. Based on the above, it can be argued that the effect of geometry on the temperature contours in the combustion chamber is significant.

Effect of geometry on H_2 mass fraction contours

Hydrogen mass fraction contour for all investigated geometries are presented in Fig. 9.

Here the hydrogen mass fraction contour for the 2D-axis geometry is represented as a mirror image along the x-axis according Fig. 1. For 3D geometry the H_2 mass fraction contour is presented as 2D contour for the XY-plane to better understand the effect of geometry. The visual comparison of the H_2

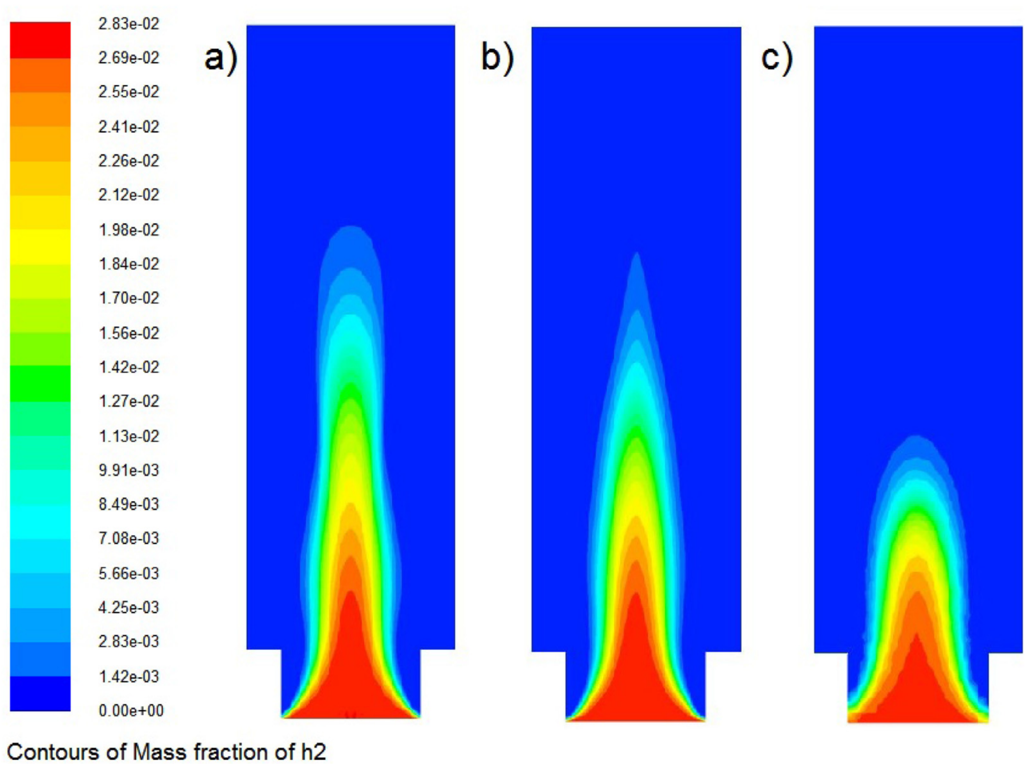

Fig. 9 – Contours of H_2 mass fraction in the combustor for different computational geometries: a) 2D-axis; b) 2D; c) 3D.

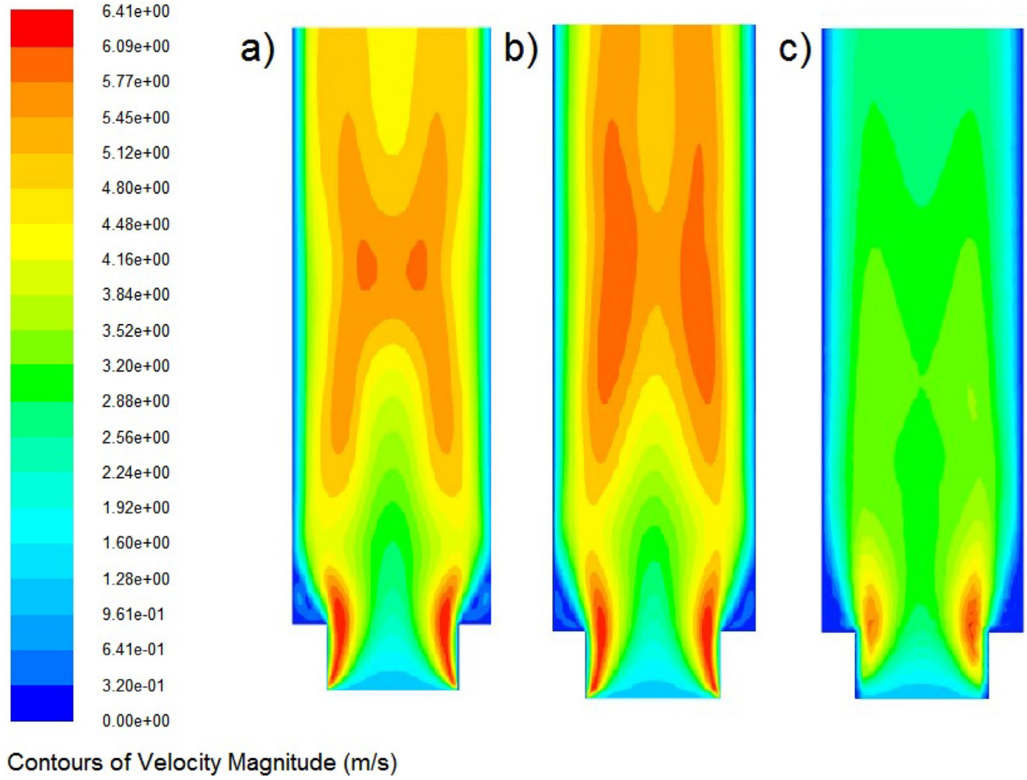

Fig. 11 – Contours of velocity in the combustor for different computational geometries: a) 2D-axis; b) 2D; c) 3D.

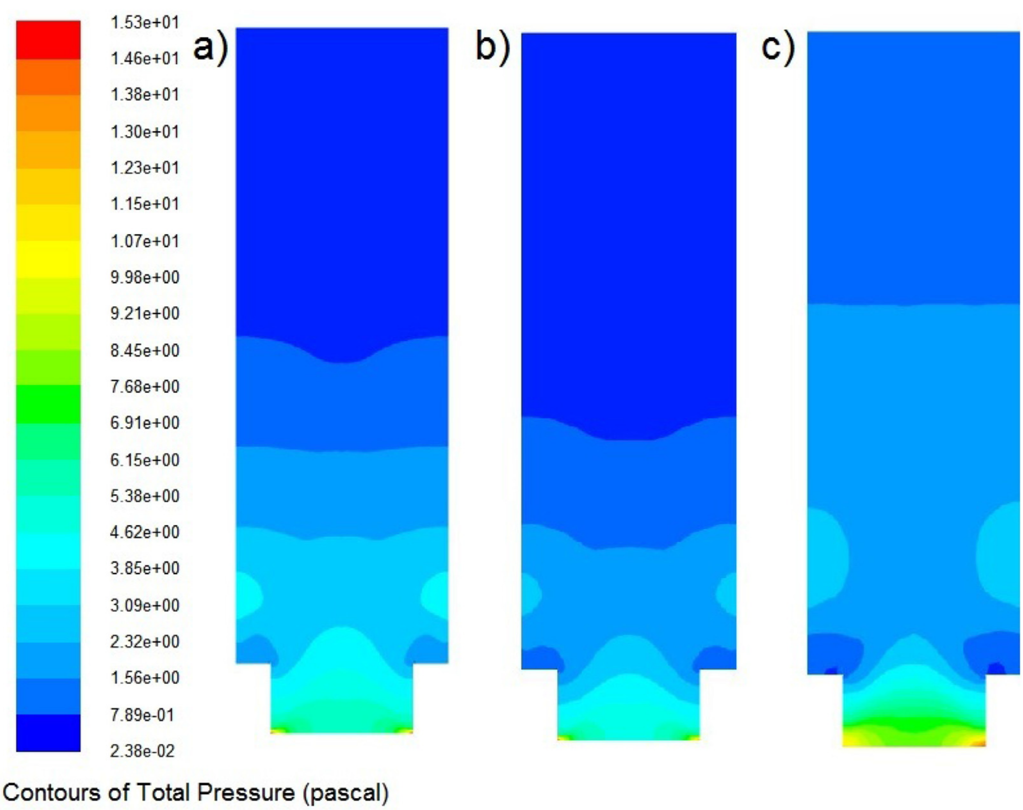

Fig. 12 – Contours of pressure in the combustor for different computational geometries: a) 2D-axis; b) 2D; c) 3D.

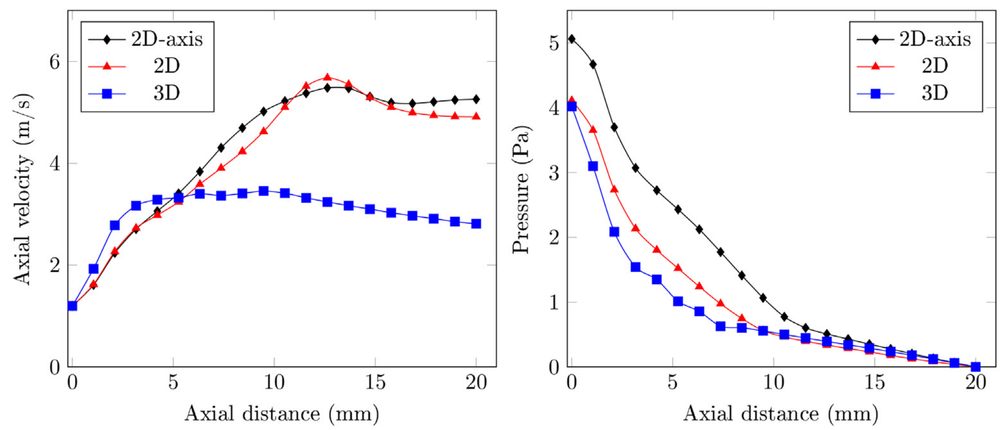

Fig. 13 – Axial velocity and pressure profiles in the combustor for different computational geometries.

mass fraction contours shows significant discrepancies in the simulation results. The maximum divergence of results occurs in the area of the flame formation. (see Fig. 10).

Hydrogen mass concentration profiles along the axial line of the investigated combustor for tested geometries are shown in Fig. 9. As seen from this figure, the near combustor inlet the hydrogen mass fraction profiles have a well agreement between results for all investigated geometries, because is no combustion in that area. In the range of the axial distance from 5 mm to 15 mm, the greatest differences are observed, because this range is the area of the flame formation. There is well agreement between the results for 3D geometry and the CFD study of Yilmaz and co-workers [34], in which the 3D geometry of micro-combustor was also used.

Effect of geometry on velocity and pressure contours

Velocity contours for all investigated fluid computational domain are depicted in Fig. 11. A visual comparison of the velocity contours shows a significant discrepancy between two-dimensional (2D-axis and 2D) and three-dimensional geometries. Also, there is a slight discrepancy between the contours for the 2D-axis and 2D geometries (Fig. 13(left side)). Such discrepancies can be explained by different temperature and hydrogen mass fraction contours for 2D-axis, 2D and 3D geometries.

As seen in Fig. 12, the contours of pressure have minimal visual differences for all the hydrogen burning characteristics that are investigated in this work. As seen in Fig. 13 (right side), axial pressure profiles for all investigated geometries (2D-axis, 2D and 3D) are almost similar in terms of trend and value. On the other hand, the divergence of the axial velocity profiles takes place for the area near the outlet between 2D and 3D geometries. The axial velocity profiles for 2D-axis and 2D geometries are almost similar in terms of trend and value.

Conclusion

In this paper, the combustion of hydrogen-air mixture was investigated by CFD-modeling of combustor for different type of geometries. The three type of geometries were analyzed (2D-axial, 2D and 3D) to investigate it effect on combustion characteristics: wall and fluid temperature, velocity, pressure and hydrogen mass fraction. The obtained results of presented numerical investigation were compared with experimental results and numerical results of other authors.

To compare the results obtained for different geometries, the number of cells for two-dimensional and three-dimensional geometry was determined, in which a further increase in their number did not have a significant effect on the final results. It is determined that for the 2D geometry (2D) the increase in the number of cells above 50,000 does not significantly effect the final result under constant convergence criteria; for 3D geometry (3D), an increase in the number of elementary cells above 400,000 does not have a significant effect on the final result under constant convergence criteria.

By comparing temperature profiles and contours obtained from numerical modeling, it was founded that the near combustor inlet the temperature profiles have a well agreement between results for all investigated geometries, because the temperature in that area is lower. However, towards the outlet of the combustor, discrepancies become significant. These difference between the temperature of the combustion products at the outlet for three-dimensional and 2D (2D and 2D-axis) geometries is more than 25%, and for the region of the flame formation, the temperatures differ by several times. It can be argued that the effect of geometry on the temperature contours in the combustion chamber is significant.

By comparing hydrogen mass fraction profiles and contours, it was concluded that the near combustor inlet the hydrogen mass fraction profiles have a well agreement between results for all investigated geometries, because is no combustion in that area. In the range of the axial distance from 5 mm to 15 mm, the greatest differences are observed, because this range is the area of the flame formation.

A visual comparison of the velocity contours shows a significant discrepancy between two-dimensional (2D-axis and 2D) and three-dimensional geometries. Also, there is a slight discrepancy between the contours for the 2D-axis and 2D geometries. Such discrepancies can be explained by different temperature and hydrogen mass fraction contours for 2D-axis, 2D and 3D geometries. The contours of pressure have minimal visual differences for all the hydrogen burning characteristics that are investigated in this work. The axial pressure profiles for all investigated geometries (2D-axis, 2D and 3D) are almost similar in terms of trend and value.

The discrepancies between results that obtained for 2D-axis, 2D-planar and 3D geometries can be explained by the following facts. Mixing and more generally diffusion phenomena, that take place at hydrogen/air combustion, are never purely 2D. 3D diffusion is accounted for by the transport equation of turbulence quantities for the used RANS model and for laminar conditions every parameters depend on geometry of computational domain. For analyzed geometry the aspect ratio (Fig. 2; $x/y \approx 7$) is sufficiently small. For such aspect ratio, the 2D model can not be considered as three-dimensional, because third dimension is not too much larger than any other dimensions. Only for large aspect ratios there should not be much difference. And 2D modeling can be considered as 3D, in which the third dimension is much large than any other dimension.

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